

Quantum Computing

Chapter 00: Introduction

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Who? Me?

- Nickname: [Arm](#) (P'/N' Arm, etc.)
- Born: Aug 1981
- Work
 - Researcher at NECTEC 2005-2024
 - Lecturer at SIIT, Thammasat University 2025-now
- Education
 - B.Eng & M.Eng in Computer Engineering, Kasetsart University, Thailand
 - Obtained Ministry of Science and Technology Scholarship of Thailand in early 2008
 - Did a PhD in Informatics (AI & Computational Linguistics) at University of Edinburgh, UK from 2008 to 2013



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Course Outline

Course Outline

First Half

- Lec 1: Introduction
- Lec 2: Complex algebra and matrices
- Lec 3: Quantum logic gates
- Lec 4: Clifford gates
- Lec 5: Non-Clifford gates
- Lec 6: State measurement
- Lec 7: Deutsch-Jozsa algorithm and phase kickback technique
- Midterm exam

Second Half

- Lec 8: Quantum noisy channels
- Lec 9: Grover's search, Fourier transform, and phase estimation
- Lec 10: Shor's algorithm
- Lec 11: Quantum simulation
- Lec 12: Variational quantum algorithms
- Lec 13: Quantum machine learning
- Lec 14: Quantum optimization
- Final exam

Grading Scheme

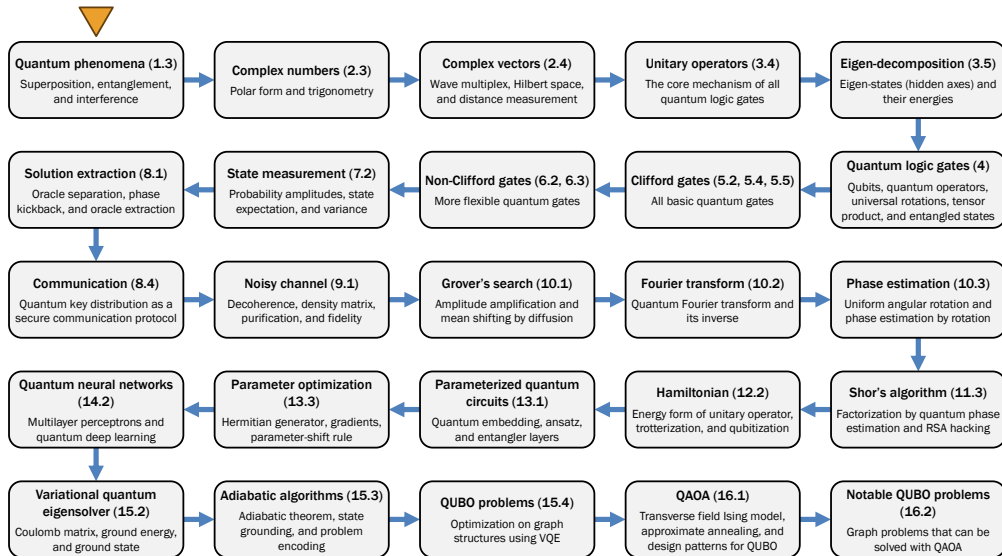
- Midterm and final exams
 - Open-slides, open-book, and open-calculator
 - Majority of my questions are based on discussion and reasoning, while only a few are calculation-oriented (i.e. virtually giving free scores)
- Assignments
 - Two take-home assignments based on calculation and quantum programming on simulators
- No-F's policy: minimum grade is **D+**
- Github repository for the course is <https://github.com/kaamanita/quantum-computing>
- Odorless foods and drinks are allowed in the class due to proximity to lunch time

Items	Proportions
Midterm	35%
Final	35%
Assignments	20%
Attendance	10%
TOTAL	100%

My Forthcoming Book

“Quantum Computing: A Gentle Introduction for AI Engineers”
(Boonkwan, 2026)

Quantum Topics You Really Need to Know



Classical vs. Quantum Computers

What the (quantum) fuss is all about?

- Hearsay: “Quantum computer is immensely faster than classical computer!”
 - Shor’s algorithm for factoring an integer n in $O((\log n)^3)$ time complexity, making RSA code decryption feasible in polynomial time
 - Quantum supremacy vs. quantum advantage
- Quantum computers
 - Perform computation by manipulating state superposition and entanglement in-memory with quantum-mechanical phenomena
 - Are susceptible to noise due to quantum interference
 - Can solve some polynomial-time and non-polynomial-time problems with bounded error
- The notion of classical computer refers to modern-day computers that rely on binary data

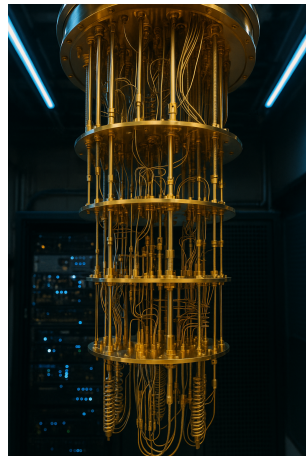


Figure 1: Quantum computer

Classical Computer

- Data: Stored as binary numbers and manipulated by electronic circuits
- Processing: Logical and arithmetic operations, data transfer, and type conversion
- Representation: Only one item out of a vast realm of all possible configurations!
- Limit: We cannot intrinsically process multiple input items in parallel

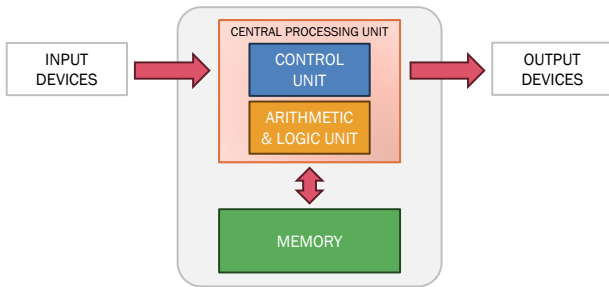


Figure 2: von Neumann architecture of classical computers

Bitstring

- Binary number is a number expressed in the base-2 numeral system: 0 and 1
- Bitstring (or binary representation) of integer N is denoted by $\mathbb{B}_L(N)$, where L is the length of such bitstring, e.g.

$$\mathbb{B}_8(33) = 00100001_2$$

- We can convert a bitstring to a decimal number by multiplying each k -th digit (from the right) with its decimal equivalent 2^{k-1} and summing these multiplicands, e.g.

$$\begin{aligned} 00100001_2 &= 1 \times 2^5 + 1 \times 2^0 \\ &= 33_{10} \end{aligned}$$

- We can also convert an integer to a bitstring by continued division as follows
 1. Find the closest exponent 2^k to N
 2. Subtract 2^k from N
 3. Repeat until the subtraction becomes zero
- Example: We convert 30_{10} to 11110_2 by

$$\begin{aligned} 30 &= 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 \\ &\quad + 1 \times 2^1 + 0 \times 2^0 \end{aligned}$$

2	30	14	6	2	0
	-16	-8	-4	-2	
	2^4	2^3	2^2	2^1	

Table 1: Conversion from 30_{10} to 11110_2

Logical Operations

- AND returns 1 iff both operands are 1

$$0011 \wedge 0101 = 0001$$

- OR returns 1 iff at least one operand is 1

$$0011 \vee 0101 = 0111$$

- NOT returns the opposite value

$$\overline{01} = 10$$

- XOR returns 1 iff only one operand is 1

$$0011 \oplus 0101 = 0110$$

- These abstract logical operations lay the foundation of data processing in classical computing, where conditional branching, jumping, and looping are predominant
- In quantum computing, these operations will be emulated by linear transformation

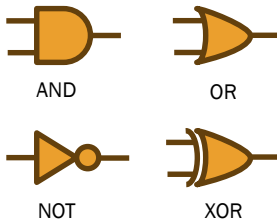


Figure 3: Logical operations

Limitations of Classical Computing

- Classical computing is linearly scalable, due to sequential logical processing of discrete states
- Parallelism in classical computing accelerates sequential processing with multiple computing cores

```
import multiprocessing as mp

def factorial(x):
    result = 1
    for i in range(2, x+1):
        result *= i
    return result

items = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
pool = mp.Pool(processes=4)
results = pool.map(factorial, items)
print(results)
```

- Classical parallelism struggles on looping with exponential iterations, resulting in $O(2^N)$ time complexity
- In quantum computing, multiple discrete values are represented as a superposition of bitstrings

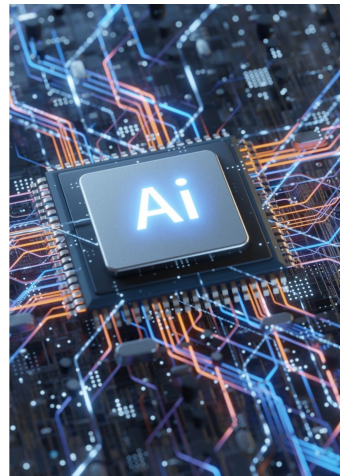
$$\{0000, 0001, \dots, 1111\} \Rightarrow \{0,1\}^4$$

Then the quantum operations are done on this state superposition

- The quantum data processing is based on complex algebra (i.e. complex vectors, complex matrices, and measurement)

Modern-Day AI

- High demand of parallel computation with graphical processing units (GPUs)
- Limitations of high performance computing
 - **Power wall:** Performance gain of transistor-based CPUs are limited by thermal ceiling
 - **Memory wall:** High bandwidth memory still lags behind when compared to the CPUs
 - **Network bottleneck:** Data synchronization between computing units are relatively very slow
 - **Quantum tunneling:** Electrons start to leak through insulation across transistors causing errors & heat



Quantum Computing

- Nondeterminism: probabilistic information processing via quantum mechanics (but we don't have to deal with it)

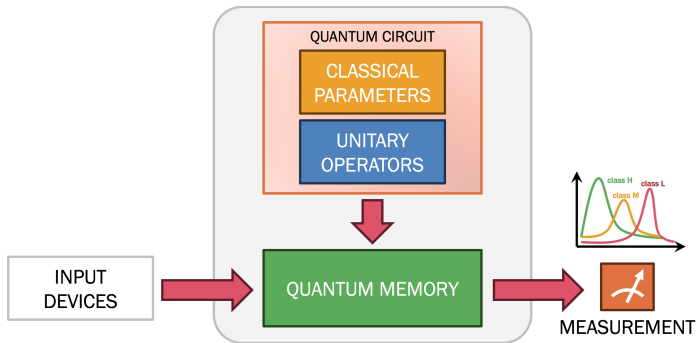


Figure 4: Quantum architecture

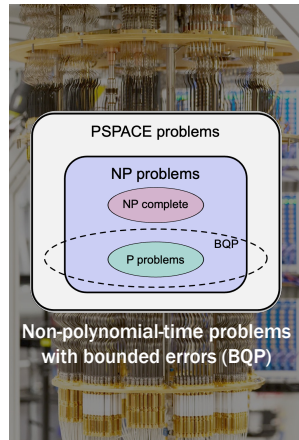


Figure 5: NP-time problems with bounded errors (BQP)

Quantum Mechanics

Quantum Memory

- Qubit: Each bit of information is stored as the momentum spin of a particle
- Information processing is to manipulate these qubits via physical interaction e.g. laser beam (trapped ions) or light interference (photons)

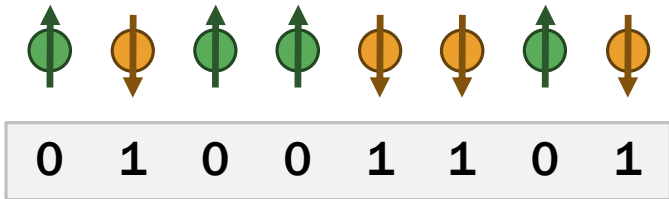


Figure 6: A classical bitstring represents the system's current state.

State Measurement

- Momentum spins are destroyed when we read the system's current state

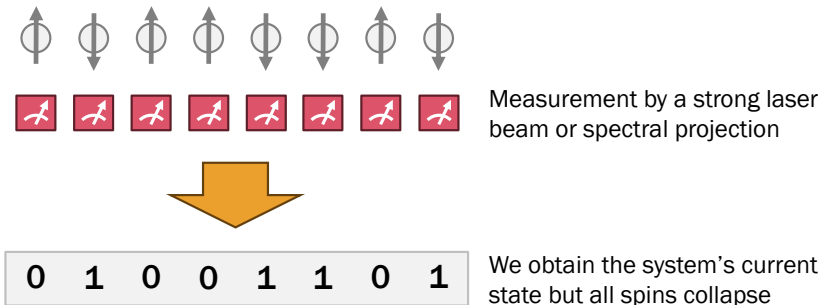


Figure 7: State measurement

Phenomenon 1: Superposition

- A quantum system can exist in multiple states (with probabilities) in parallel

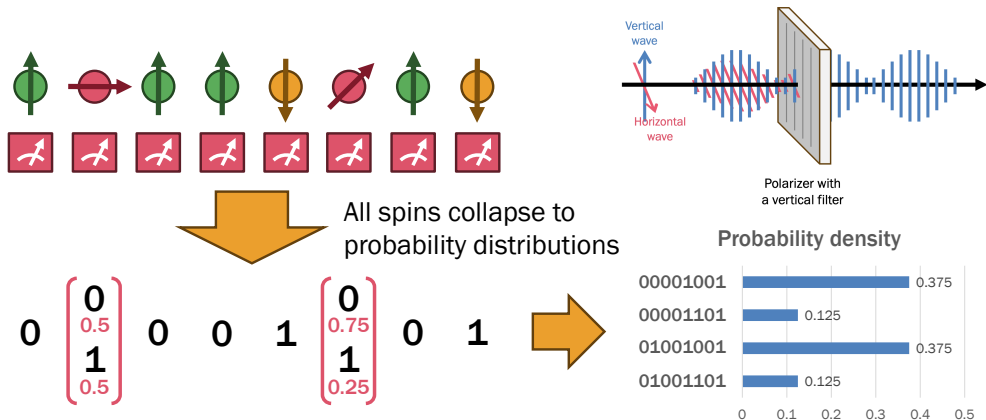


Figure 8: Quantum superposition

Quantum State and Probability Amplitudes

- Quantum state: the system's configuration of momentum spins for each qubit
- Three dipoles (0/1, +/−, and +i/−i) represent the 3D momentum spin of one qubit

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

where complex numbers a and b are probability amplitudes of $|\psi\rangle$ collapsing to 0 and 1

- Probability density of $|\psi\rangle$ collapsing to 0 or 1:

$$P(0|\psi) = a^2$$

$$P(1|\psi) = b^2$$

- Dirac's notation: $|\psi\rangle$ reads 'ket'-psi

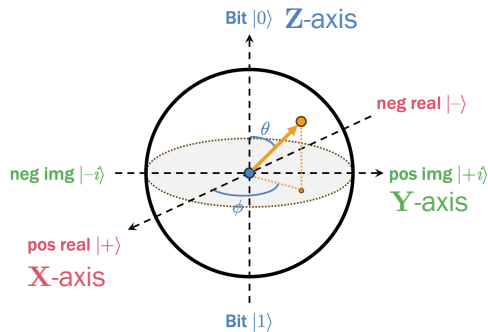


Figure 9: Bloch sphere

Multi-Qubit System

- Tensor product of quantum states

$$\begin{aligned} & (a_1|0\rangle + b_1|1\rangle) \otimes (a_2|0\rangle + b_2|1\rangle) \\ &= a_1a_2|00\rangle + a_1b_2|01\rangle + b_1a_2|10\rangle + b_1b_2|11\rangle \end{aligned}$$

where $\otimes$ denotes concatenation

- Factorization of quantum state

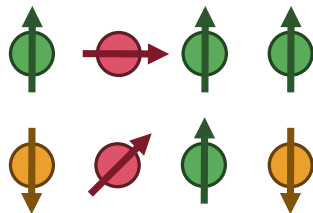
$$\begin{aligned} & \frac{\sqrt{3}}{2\sqrt{2}}|00001001\rangle + \frac{1}{2\sqrt{2}}|00001101\rangle \\ & + \frac{\sqrt{3}}{2\sqrt{2}}|01001001\rangle + \frac{1}{2\sqrt{2}}|01001101\rangle \end{aligned}$$

Multi-state superposition



$$\begin{aligned} & |0\rangle \otimes \left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \right) \otimes |0\rangle \otimes |0\rangle \\ & \otimes |1\rangle \otimes \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle \right) \otimes |0\rangle \otimes |1\rangle \end{aligned}$$

Tensor product



Dirac's Notation

- Single-qubit ket = 2-dim column vector

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

- L -qubit ket = 2^L -dim column vector

$$|k\rangle = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \quad \sum_{k=0}^{2^L-1} a_k |k\rangle = \begin{bmatrix} a_0 \\ \vdots \\ a_{2^L-1} \end{bmatrix}$$

- $|k\rangle$ is 1-hot vector at the k -th position

- Single-qubit bra = 2-dim row vector

$$\begin{aligned} (a|0\rangle + b|1\rangle)^* &= \bar{a}\langle 0| + \bar{b}\langle 1| \\ &= \begin{bmatrix} \bar{a} & \bar{b} \end{bmatrix} \end{aligned}$$

- L -qubit bra = 2^L -dim row vector

$$\sum_{k=0}^{2^L-1} a_k \langle k| = \begin{bmatrix} a_0 & \dots & a_{2^L-1} \end{bmatrix}$$

- Bra-ket product = vector similarity

$$\langle k|k\rangle = 1 \quad \langle j|k\rangle = 0$$

where $j \neq k$

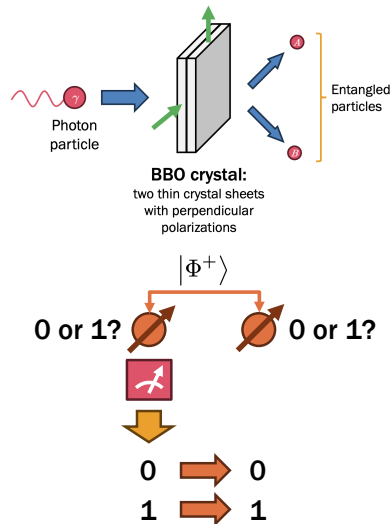
Phenomenon 2: Entanglement

- Multiple particles may become inextricably linked to each other, i.e. they are entangled
- Entangled states are multi-qubit states that cannot be factorized further, e.g. Bell states

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}$$

- If one of them collapses, the rest of the pack also collapse instantaneously:

$$P(00|\Phi^+) = \frac{1}{2} \quad P(11|\Phi^+) = \frac{1}{2}$$



Phenomenon 3: Quantum Interference

- If we manipulate the prob amplitude of a qubit, this change ripples to the others
- Probability amplitude can be manipulated in two ways
 - **Amplification:** increasing the prob amplitude
 - **Dampening:** decreasing the prob amplitude
- Quantum interference is the key mechanism of quantum gate design

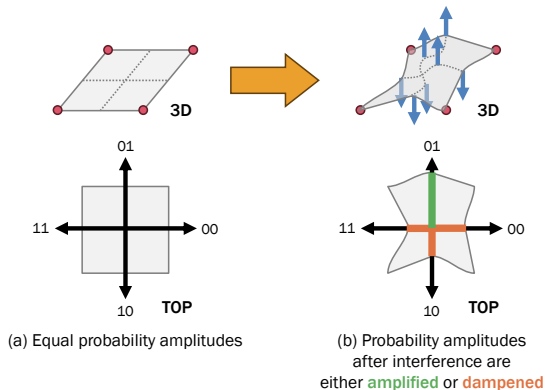


Figure 10: Quantum interference

Phenomenon 4: Quantum Decoherence

- Quantum state loses its property due to an unexpected interaction with environment
- Noisy qubit is denoted by a density matrix (comparable to covariance matrix in ML)
- Pure state:

$$\begin{aligned}\rho(\psi) &= |\psi\rangle \langle \psi| \\ &= \begin{bmatrix} a \\ b \end{bmatrix} \begin{bmatrix} a^* & b^* \end{bmatrix} \\ &= \begin{bmatrix} |a|^2 & a\bar{b} \\ \bar{a}b & |b|^2 \end{bmatrix}\end{aligned}$$

where purity is $\text{tr}[\rho(\psi)] = |a|^2 + |b|^2 = 1$

- Mixed state: is a noisy qubit

$$\rho(\psi) = \begin{bmatrix} a_{00} & \overline{a_{01}} \\ a_{01} & a_{11} \end{bmatrix}$$

where purity is $\text{tr}[\rho(\psi)] = a_{00} + a_{11} \neq 1$

- Fidelity: similarity between two quantum states (usually between a mixed state and its target pure state)

$$F(\rho, \sigma) = \left[\text{tr} \sqrt{\sqrt{\rho} \sigma \sqrt{\rho}} \right]^2$$

- We measure the quality of quantum information storage in terms of fidelity: the state-of-the-art fidelity is 99.5-99.7%

Quantum Operations

Designing a Quantum Operator/1

- We manipulate the system's quantum state by applying a quantum operator
- [State mapping](#)

$$\begin{array}{c} |\text{input } \psi_1\rangle \mapsto |\text{output } \phi_1\rangle \\ \vdots \\ |\text{input } \psi_K\rangle \mapsto |\text{output } \phi_K\rangle \end{array}$$

- [Quantum operator](#)

$$U = \sum_{k=1}^K |\text{output } \phi_k\rangle \langle \text{input } \psi_k|$$

- Reminder: $\langle \psi| = |\psi\rangle^*$

- Generally, we use pure states as inputs ψ_k
- The quantum operator applies [interference](#) on the target qubits

$$\begin{aligned} & U |\text{state } \xi\rangle \\ &= \left(\sum_{k=1}^K |\text{output } \phi_k\rangle \langle \text{input } \psi_k| \right) |\text{state } \xi\rangle \\ &= \sum_{k=1}^K |\text{output } \phi_k\rangle \underbrace{\left(\langle \text{input } \psi_k| |\text{state } \xi\rangle \right)}_{\text{similarity}} \\ &= \sum_{k=1}^K |\text{output } \phi_k\rangle \langle \text{input } \psi_k| \text{state } \xi \rangle \end{aligned}$$

Designing a Quantum Operator/2

- Example: NOT operator has state mapping

$$|0\rangle \mapsto |1\rangle \quad |1\rangle \mapsto |0\rangle$$

The quantum operator for NOT is

$$\begin{aligned} \mathbf{X} &= |1\rangle\langle 0| + |0\rangle\langle 1| \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \end{aligned}$$

- Compared with matrix multiplication

$$\begin{aligned} \mathbf{X} \begin{bmatrix} a \\ b \end{bmatrix} &= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} \\ &= \begin{bmatrix} 0a + 1b \\ 1a + 0b \end{bmatrix} \\ &= \begin{bmatrix} b \\ a \end{bmatrix} \end{aligned}$$

- Compared with state mapping

$$\begin{aligned} \mathbf{X}(a|0\rangle + b|1\rangle) &= a(\mathbf{X}|0\rangle) + b(\mathbf{X}|1\rangle) \\ &= a|1\rangle + b|0\rangle \\ &= b|0\rangle + a|1\rangle \end{aligned}$$

NOT Gate: X

- X gate flips the qubit between $|0\rangle$ and $|1\rangle$
- State mapping

$$|0\rangle \mapsto |1\rangle \quad |1\rangle \mapsto |0\rangle$$

- It imitates logical NOT in classical computing

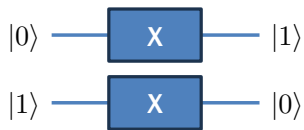
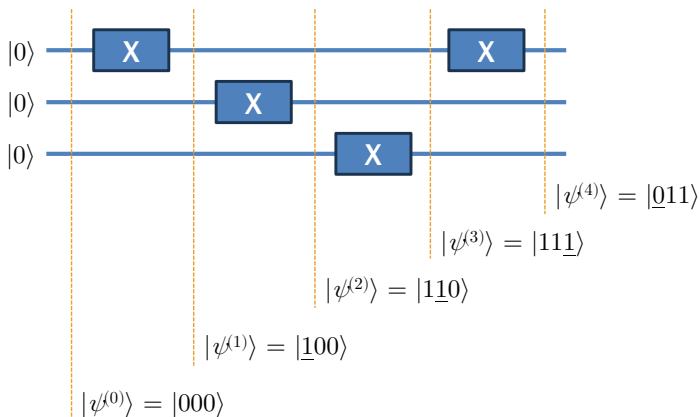


Figure 11: X gate

- Quantum algorithm is a circuit of quantum logic gates



Hadamard Gate: H

- H gate flips between pure states and superpositions

- State mapping

$$|0\rangle \mapsto \underbrace{\frac{|0\rangle + |1\rangle}{\sqrt{2}}}_{|+\rangle} \quad |1\rangle \mapsto \underbrace{\frac{|0\rangle - |1\rangle}{\sqrt{2}}}_{|-\rangle}$$

- Used for parallel processing

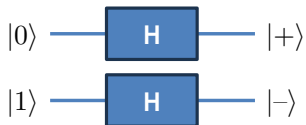
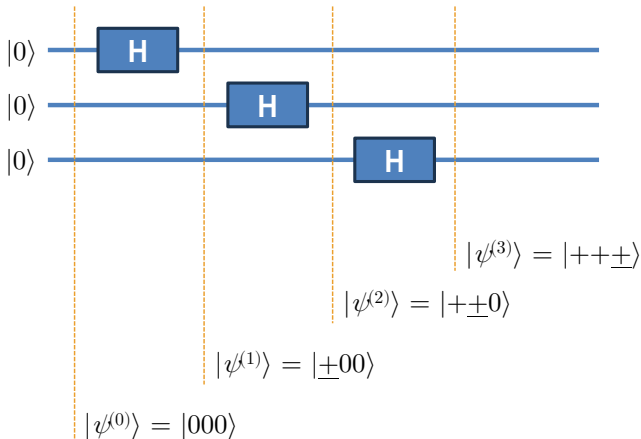


Figure 12: Hadamard gate

- Preparing a superposition of all possible 2^L inputs



Controlled NOT Gate: CNOT

- CNOT gate imitates an if-clause and creates entanglement
- State mapping

$$|0x\rangle \mapsto |0x\rangle \quad |1x\rangle \mapsto |1\bar{x}\rangle$$

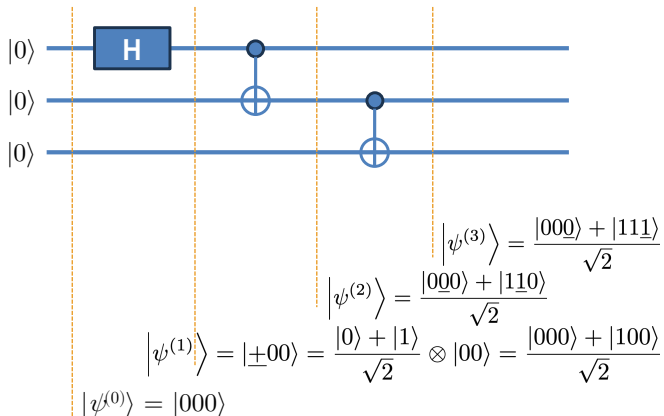
where x is either 0 or 1

- Last qubit is the output equivalent to classical XOR gate



Figure 13: Controlled NOT gate

- Preparing an entangled state with H and CNOT gates



Phase Shift Gate: S

- S_gate flips between real and complex numbers on $|1\rangle$
- State mapping

$$|0\rangle \mapsto |0\rangle \quad |1\rangle \mapsto i|1\rangle$$

- Imaginary number $i = \sqrt{-1}$ is used as soft feature in quantum machine learning

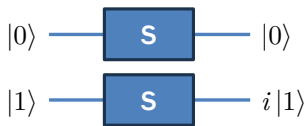
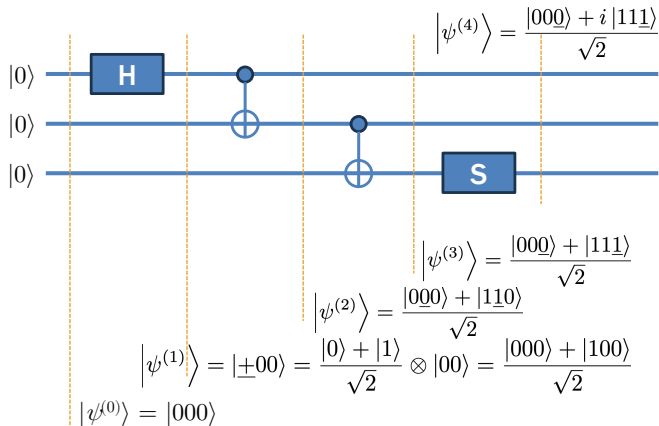


Figure 14: Phase shift gate

- Preparing an entangled state with complex probability amplitudes



Toffoli Gate: CCNOT

- CCNOT gate imitates an if-clause with two control qubits
- State mapping

Both controls: $|11x\rangle \mapsto |11\bar{x}\rangle$

Otherwise: $|xyz\rangle \mapsto |xyz\rangle$

where x and y are binary flags

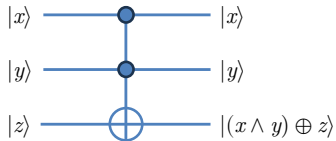
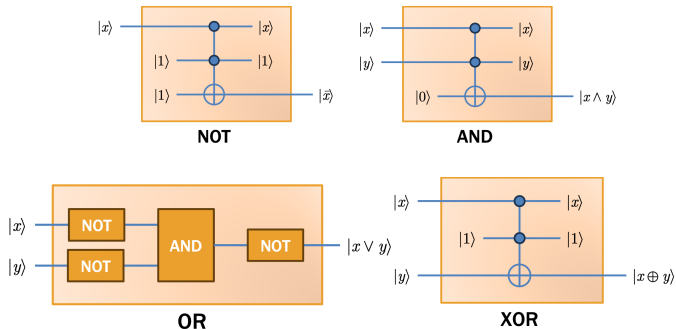


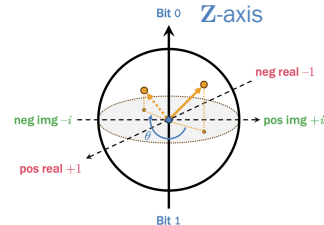
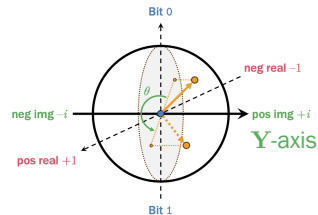
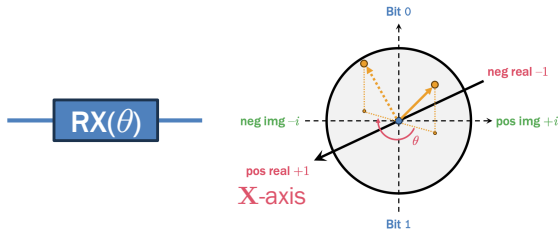
Figure 15: Toffoli gate

- CCNOT gate imitates all classical logic gates, allowing massive parallelism on classical programs



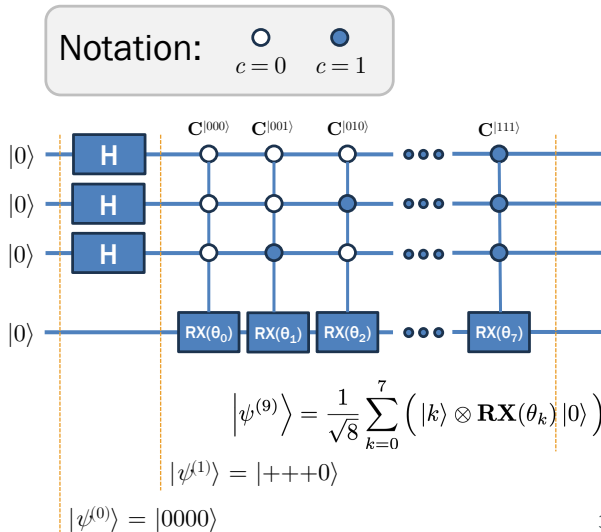
Rotation Gates: R_X , R_Y , and R_Z

- Rotation gates rotate a qubit around the specified eigenbases
- Rotation alters the probability densities used in quantum machine learning
- X-rotation $R_X(\theta)$, Y-rotation $R_Y(\theta)$, and Z-rotation $R_Z(\theta)$, where θ is a model parameter



Controlled Gate: $C^{|k\rangle}U$

- Extension of CNOT, where control qubits are $|k\rangle$
- State mapping of $C^{|k\rangle}U$
 - If controls $|x_1, \dots, x_L\rangle = |k\rangle$
$$|x_1, \dots, x_L, y\rangle \mapsto |x_1, \dots, x_L\rangle \otimes U|y\rangle$$
 - Otherwise
$$|x_1, \dots, x_L, y\rangle \mapsto |x_1, \dots, x_L, y\rangle$$
- This is frequently used in Fourier transform and phase estimation

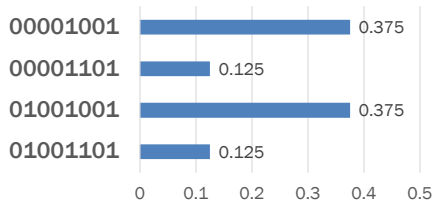


State Measurement

- Probability density: We destructively measure the probability distribution of each pure state by projection (i.e. the quantum state collapses)



Probability density



- Expectation: We non-destructively measure if each k -th qubit leans towards one side of the particular dipoles $\mathbf{U} \in \{\mathbf{X}, \mathbf{Y}, \mathbf{Z}\}$

$$\langle \mathbf{U}^{(k)} \rangle = \langle \psi | \mathbf{U} | \psi \rangle$$



Expectations	Positive (+1)	Negative (-1)
$\langle \mathbf{Z}^{(k)} \rangle$ or Z_k	Bit 0	Bit 1
$\langle \mathbf{X}^{(k)} \rangle$ or X_k	Pos real	Neg real
$\langle \mathbf{Y}^{(k)} \rangle$ or Y_k	Pos img	Neg img

Quantum Algorithms

Oracle Extraction

- Bernstein-Vazirani algorithm
 - Extracting an oracle $\mathbf{s}$ from a black-box function f_s by converting it to U_{f_s}
- Phase kickback technique
 - State mapping of U_{f_s}

Oracle: $|s_{1:N}, y\rangle \mapsto |s_{1:N}, y \text{ cis } \theta\rangle$

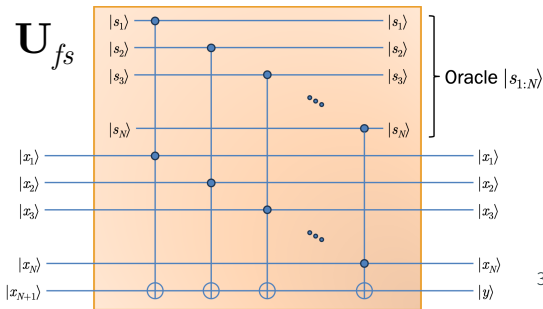
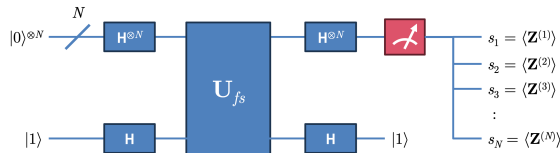
Otw.: $|x_{1:N}, y\rangle \mapsto |x_{1:N}, y\rangle$

- Kickback property

$$U_{f_s} \left(|+\rangle^{\otimes N} \otimes |-\rangle \right) = \left(\bigotimes_{k=1}^N \frac{|0\rangle + (\text{cis } \theta)^{f_{sk}(2^{k-1})} |1\rangle}{\sqrt{2}} \right) \otimes |-\rangle$$

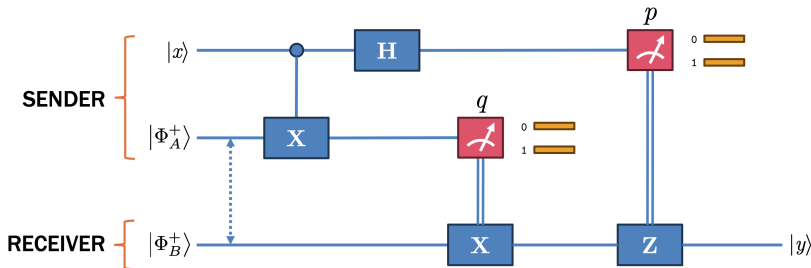
- The last qubit interferes the others

The pure state $\mathbf{s}$ will have the highest probability density



Quantum Teleportation

- Secure communication based on quantum mechanics
- We want to transmit state $|x\rangle = a|0\rangle + b|1\rangle$ with entangled state $\Phi^+ = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$
- How it works:
 - The sender and receiver have $|\Phi_A^+\rangle$ and $|\Phi_B^+\rangle$, respectively
 - The sender runs the quantum teleportation algorithm on the input qubit $|x\rangle$
 - The sender measures p and q and send them over classical communication
 - The receiver uses them to translate $|\Phi_B^+\rangle$ to $|x\rangle$



p	q	State $ y\rangle$
0	0	$a 0\rangle + b 1\rangle$
0	1	$b 0\rangle + a 1\rangle$
1	0	$a 0\rangle - b 1\rangle$
1	1	$-b 0\rangle + a 1\rangle$

Quantum Key Distribution

- Secure communication without actual key transmission, such as [BB84](#)

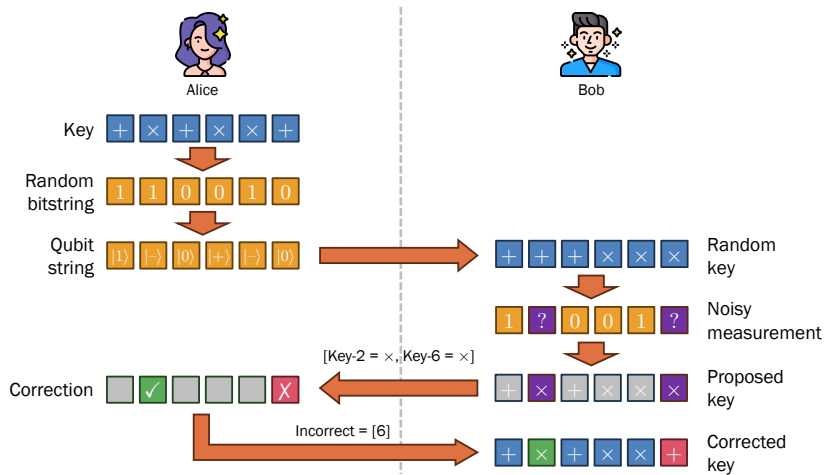


Figure 16: BB84 Protocol

Grover's Search Algorithm

- We can search for 1 of N items w.r.t. the oracle function f with $k^* = \left\lfloor \frac{\pi}{4} \sqrt{N} \right\rfloor$ iterations

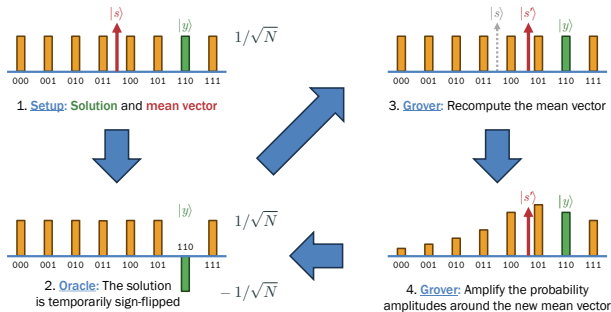


Figure 17: Overview of Grover's algorithm

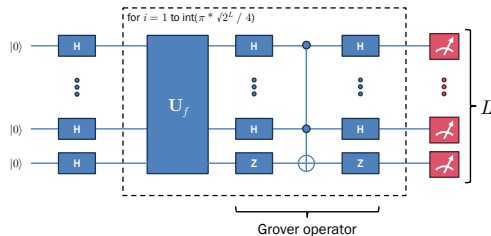
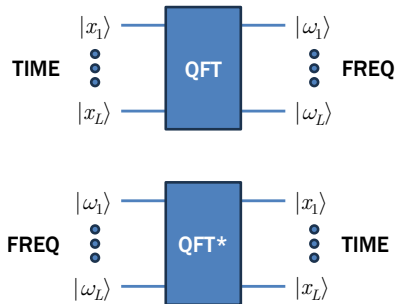
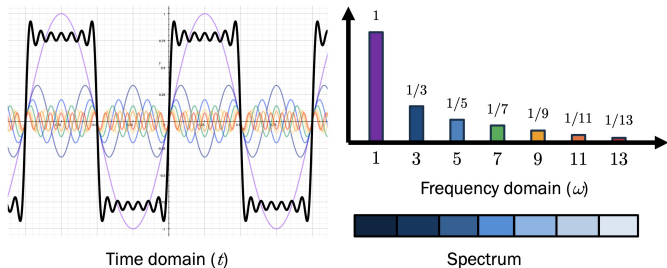


Figure 18: Grover's algorithm. U_f is equivalent to Step 2, while the Grover's operator is equivalent to Steps 3 and 4.

Quantum Fourier Transform

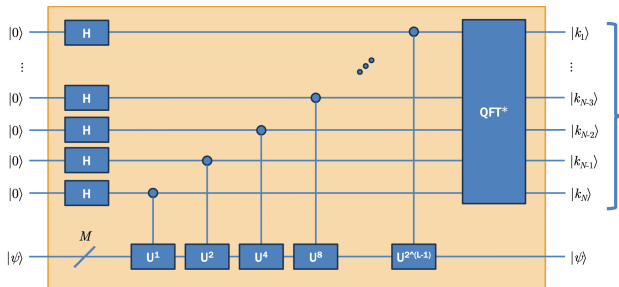
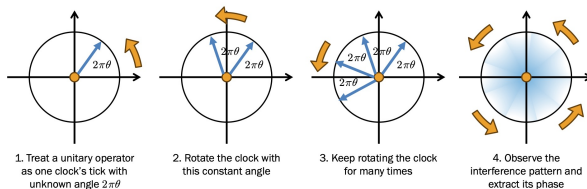
- Translates a time series of N items to a frequency domain, and vice versa, with $O((\log N)^2)$ time complexity



where $L = \log N$

Quantum Phase Estimation

We can extract a phase $2\pi\theta$ of a black-box algorithm $\mathbf{U}$ by rotating it in different frequencies and observe the resonance frequency



$\text{QPE}_N(\mathbf{U})$

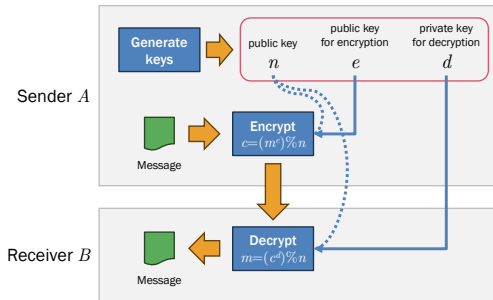
Measure for $k^* = \text{pure state with max prob}$

Phase $\theta = k^*/2^N$

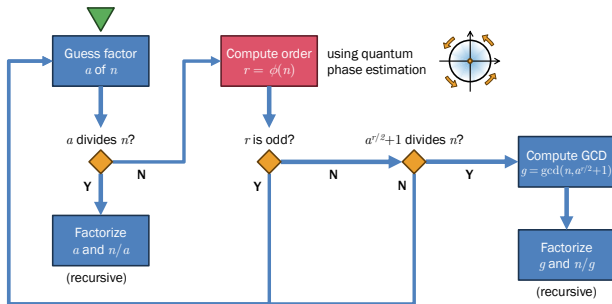
Quantum-Suitable Problems

Shor's Algorithm

- RSA cryptosystem
 - Public and two private keys (n, e, d)
 - Encryption: $c = (m^e) \% n$
 - Decryption: $m = (c^d) \% n$



- Attack with QPE
 - Modulo $\text{Umod } n(a)$ is eqv. to rotation
 - QPE is used to guess the phase
 - Time complexity $O(L^2 \log L)$



Simulation with Trotterization

- Hamiltonian $\hat{H}_U$ is the energy form of quantum circuit U of L qubits

$$U = \text{cis}(-\hat{H}_U)$$

- We can compute $\hat{H}_U$ by

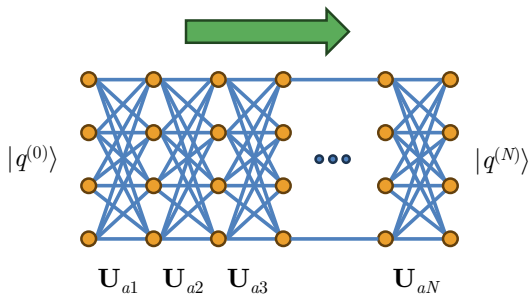
$$\hat{H}_U = \sum_{P \in \{I, X, Y, Z\}^{\otimes L}} [P \times \text{trace}(UP)]$$

where I, X, Y, Z are Pauli matrices

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

- Trotter-Suzuki decomposition: We can simulate N iterations of U by approximation

$$U^N \approx \lim_{n \rightarrow \infty} \left[\text{cis} \left(-\frac{N}{n} \hat{H}_U \right) \right]^n$$



- Variational quantum algorithm

- Quantum circuit with rotation gates $\mathbf{RX}(\theta)$, $\mathbf{RY}(\theta)$, and $\mathbf{RZ}(\theta)$
- Basic entangler layer (**BEL**)

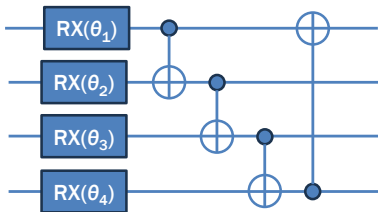


Figure 19: Basic entangler layer

- Quantum embedding $\Lambda(\Theta)$
 - Angle embedding: params for $\mathbf{RX}(\theta_k)$
 - Amplitude embedding: params for each pure state
- Hybrid classical-quantum computing

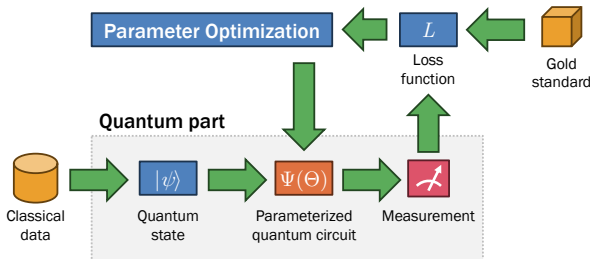
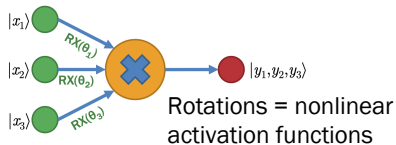


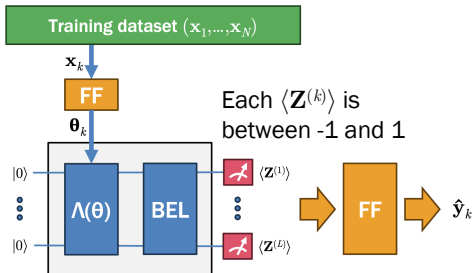
Figure 20: Hybrid system

Quantum Neural Networks

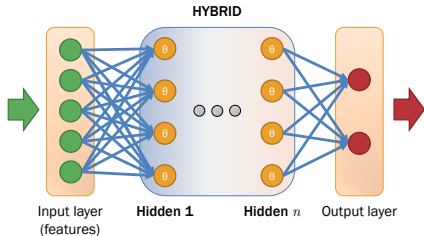
- Quantum perceptron



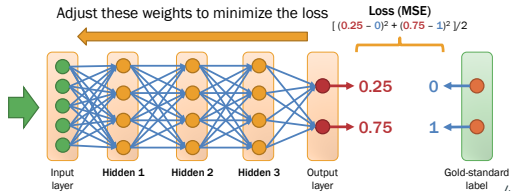
- Quantum multilayer perceptrons



- Quantum deep NNs



- Backpropagation



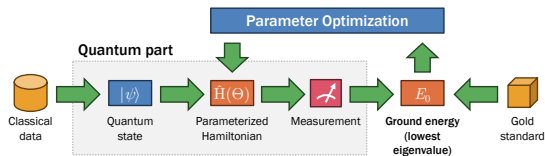
Variational Quantum Eigensolver (VQE)

- We want to minimize a loss Hamiltonian defined on qubit expectations

$$L(\Theta) = f\left(\langle \mathbf{Z}^{(\psi_1)} \rangle, \dots, \langle \mathbf{Z}^{(\psi_L)} \rangle\right)$$

- Parameter shift rule: Computing gradient for SGD by shifting Θ by $\pi/2$

$$\frac{\partial}{\partial \Theta} L(\Theta) = \frac{1}{2} \left(L\left(\Theta + \frac{\pi}{2}\right) - L\left(\Theta - \frac{\pi}{2}\right) \right)$$



- NFT Algorithm: Gradually re-estimating each parameter at timestep t

$$\theta_j^{(t+1)} = -\frac{\pi}{2} - \arctan \frac{y_j^{(t)} - z_j^{(t)}}{2x_j^{(t)} - y_j^{(t)} - z_j^{(t)}}$$

where

$$x_j^{(t)} = L(\Theta^{(t)} | \theta_j = \theta_j^{(t)})$$

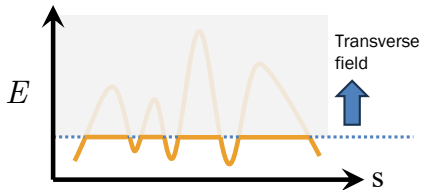
$$y_j^{(t)} = L(\Theta^{(t)} | \theta_j = \theta_j^{(t)} + \frac{\pi}{2})$$

$$z_j^{(t)} = L(\Theta^{(t)} | \theta_j = \theta_j^{(t)} - \frac{\pi}{2})$$

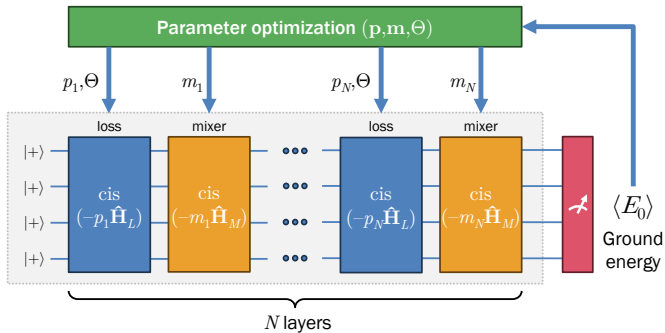
- Both methods converge very fast

Quantum Approximate Optimization Algorithm (QAOA)

- Quantum annealing consists of two steps
 - **Mixer**: explore the transverse field to improve coverage
 - **Loss**: identify the globally optimal point in the explored transverse field



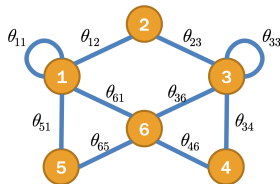
- Objective function = N layers of mixer Hamiltonian $\hat{H}_M$ + loss Hamiltonian $\hat{H}_L$



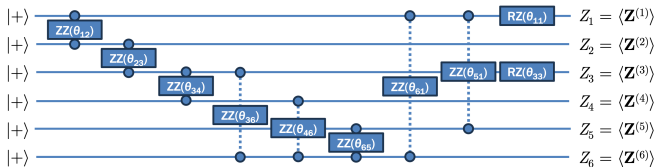
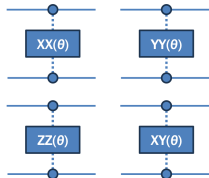
QAOA: Quadratic Unbounded Binary Optimization (QUBO)

- Each edge (i, j) becomes a model parameter θ_{ij}
- Loss function:

$$L = \sum_{(i,i) \in E} \frac{\theta_{ii}}{2} Z_i + \sum_{(i,j) \in E} \frac{\theta_{ij}}{2} Z_i Z_j$$

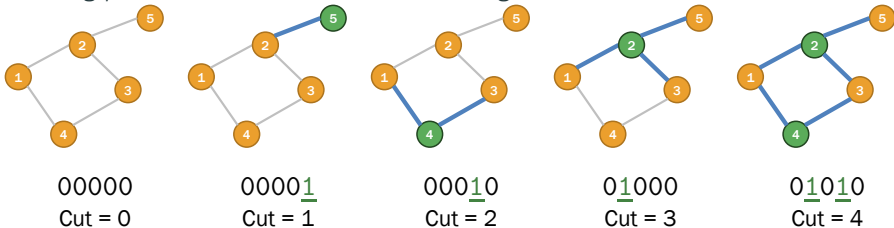


Ising gates rotate both input qubits with the same phase θ



QAOA: Max-Cut as a QUBO Problem

- Graph-coloring problem where the number of edges between two colors is maximum



- Mixer Hamiltonian: X-mixer applies bit-flipping between 0 and 1 on each qubit

$$\hat{H}_M = \sum_{j=1}^N X_j$$

- Loss Hamiltonian: The more adjacent nodes of different colors, the less loss

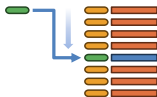
$$\hat{H}_L = \sum_{i,j \in E} Z_i Z_j$$

Conclusion

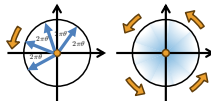
What Problems are Suitable for Quantum Computers?



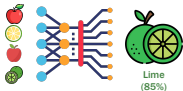
Extreme parallelization



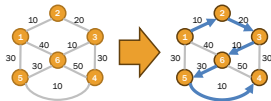
Information search



Phase estimation



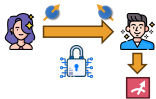
Deep learning with
adaptive activation functions



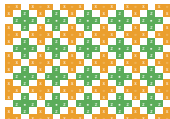
Graph-based optimization



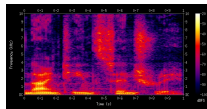
Parallelizable simulation



Secure communication
and cryptography



Error correction



Spectral analysis

- Superposition allows quantum computers to perform independent tasks at the same time
- Entanglement fortifies the security of data communication
- The output will be a probability distribution of possible outputs
- **Warning:** Quantum computers work on complex matrices and eigen-decomposition

Figure 21: Quantum-suitable problems

Questions?